# Confidential mode v1 vs v2: who can ever see the node's API key

**The short version.** Both versions hide prompts and answers from the node, pin the model and token cap, and bill from the usage the provider reports. They differ in one thing: **v1** keeps the node's provider API key from the client only as long as the client never sees the node's own encrypted traffic to the provider. **v2** removes that condition: client and node compute the TLS handshake keys jointly (a two-party computation, 2PC), so the client never holds the key that encrypts the record carrying the API key, even if it records the node's traffic. The price is bandwidth: about 21 MB each way per request, sent before the request exists.

| | **v1** (shipping) | **v2** (two-party handshake) |
|---|---|---|
| Node reads prompt or answer | no | no |
| Client can recover the node's API key | yes, **if** it (or anyone it colludes with) sees the node→provider ciphertext: a tap on the node's uplink, its hosting provider, a CDN edge | **no**: the record is encrypted under a key only the node has; recording it gives nothing |
| Model / token cap / billing | pinned by proofs, provider-reported usage | same |
| Who holds which TLS secret | client: all of them | client: handshake, response and body keys; node: handshake keys and the one key for its own record; nobody: the master secret |
| Key exchange | X25519 | P-256 (the joint key share needs a full curve point) |
| Data per request, client and node (both directions) | about 6 MB | about 46 MB (42 MB 2PC tables, 4 MB proofs) |
| Node upload per request | about 3 MB | about 23 MB (100 requests/min about 0.3 Gbit/s) |
| New cryptography in the trust base | ZK proofs | ZK proofs **and** a 2PC (dual execution, share conversion), not yet externally reviewed |

**Speed.** Measured with real Venice (`deepseek-v4-flash`, 40 tokens), client and node on one host, medians of 5 sessions each:

| | v1 | v2 |
|---|---|---|
| Cold: request start → body sent | 0.73 s | 0.89 s (2PC 0.55 s, of which tables 0.49 s; π_NK 0.17 s) |
| Pre-warmed: claim → body sent | 3 ms | 5 ms |
| Suffix proof π_C2 (before the provider may start) | 0.45 s | **0.10 s** |
| Pre-warm (arm) time per session | 0.71 s | 0.88 s |

Across a 50 ms network link (from the component benchmarks): v2's tables take **1.8 s at 100 Mbit/s and 8.6 s at 20 Mbit/s**, its online 2PC about 0.45 s; in exchange π_C1 (1.5 s) disappears and π_C2 drops from 1.5 s to 0.8 s. Cold, v2 roughly matches v1 on a fast link and is several seconds slower on a slow uplink. **Pre-warmed, v2 is faster** (only the suffix proof is on the path, and it is half the cost), so v2 wants pre-warm: the tables cannot be reused, one set per request, but they are input-independent and can be built and sent ahead.

**Security for the client (prompt and answer).** v1: nothing new; the client alone runs standard TLS. v2: the client still verifies the provider's certificate itself and alone holds the body and response keys, but those keys now come out of a 2PC with the node. The 2PC is built so that a cheating node is caught before any key is used (each side garbles once and the outputs are compared); the most a cheating node can learn is **one bit** about a fresh, single-use key share, and the session then aborts. The node can write more than one record in its epoch (v1 gave it material for exactly one), but that epoch carries none of the client's data, and in v1 the client could not see the node's wire bytes either.

**Security for the node (API key).** v1: the key is safe only on a trusted egress path; one leak exposes it to anyone until rotated. v2: safe against a client that records everything; a cheating client also learns at most one bit about a single-use share, and cannot substitute its own key share (the node relays only a ClientHello carrying the joint share). v2 reveals nothing extra to the client and drops π_C1. In both versions the node must trust the provider with its key, as it does today.

**Choosing.** v1 for nodes whose upstream path is under their control and for low-bandwidth clients without pre-warm. v2 for nodes on shared or untrusted infrastructure, or with valuable keys: it turns P1 from an operational assumption into a cryptographic property, for about 21 MB per request in each direction. A node runs one version; clients pick it up from the signed offer.